

3.2.2

International benchmark

Healthcare Access and Quality (HAQ) Index is a measure developed to assess the performance of healthcare systems worldwide, focusing on how well a country or region provides access to and quality of healthcare services. The index ranges from 0 to 100, with higher scores indicating better access to and quality of healthcare.

India ranks 44th in the global Healthcare Access and Quality (HAQ) Index with a score of 65.4. Key improvement areas include expanding infrastructure, increasing the workforce, and enhancing service quality and accessibility, particularly in rural areas.

Top-ranked countries are Taiwan (1st, score 89.5) for AI integration and quality services, South Korea (2nd, score 88.0) for sustainable practices and technology, and Japan (3rd, score 87.8) for personalized healthcare and prevention.

Worldwide, the USA and UK are leading the way in offering advanced programs that could benefit India. The USA's initiatives include cutting-edge training, focus on telemedicine, and AI diagnostics, while the UK's NHS focuses on comprehensive training and digital health tools. Adopting these practices could enhance India's healthcare system by improving access, quality, and efficiency.^{43,44}

3.2.3

Sectoral trends

Emerging technologies in healthcare sector

The top healthcare trends and innovations include Artificial Intelligence, the Internet of Medical Things, Telemedicine, Big Data and Analytics, Immersive Technology, Mobile Health, 3D Printing, Blockchain, Cloud Computing and Genomics⁴⁵.

According to the World Economic Forum, the healthcare sector is set for significant job growth in the coming years, especially in tech and data roles which will open new opportunities for professionals with healthcare

knowledge with tech skills. The Indian healthcare sector is focusing on training its workforce in emerging technologies such as IoMT, blockchain, Telehealth, telemedicine Artificial Intelligence and Machine Learning, Big Data and Analytics and Augmented and Virtual Reality to enhance service delivery.

Workforce trends

By 2030, the demand for healthcare professionals is expected to double due to a shortage of workers. Roles such as medical and health services managers, nurse practitioners, physician assistants, home health aides and physical therapists will see the highest growth⁴⁶.

Some of the other emerging job roles also highlighted during discussion in healthcare sector are as follows:



AI and Machine Learning specialists:

These professionals develop and implement AI algorithms for healthcare applications, such as diagnostic tools and predictive models.



Telemedicine coordinators:

They oversee virtual care programs, arrange remote consultations, and facilitate clear communication between patients and healthcare providers.



Health data scientists:

These professionals analyze complex healthcare data to extract insights, enhance patient outcomes and optimize healthcare operations.



IoMT systems engineers:

They design, implement, and maintain networks of connected medical devices and ensure data security and interoperability.



Digital health product managers:

Oversee the development and implementation of digital health solutions, ensuring they meet user needs and regulatory standards. They collaborate with stakeholders to drive product strategy, manage roadmaps, and optimize user experience.



Robot-assisted surgery technicians:

Surgical professionals assist surgeons in operating AI-powered robots and manage the technical aspects of robotic surgical systems during procedures.

3.2.4

Future of jobs: insights from industry leaders

Emerging job roles in healthcare

The healthcare sector is poised for substantial transformation, leading to the creation of several new job roles. Our survey reveals the following job roles are anticipated to emerge in the near future:



Health data analyst: This role is expected to be in high demand, with 50% of respondents identifying it as a critical job role for the future. The increasing reliance on data-driven decision-making in healthcare underscores the need for professionals capable of analyzing vast datasets to improve patient outcomes and operational efficiency.



Genomic counsellor: Another emerging role, highlighted by 50% of respondents, is that of a genomic counsellor. As personalized medicine and genomics gain traction, the demand for specialized guidance based on genetic information will grow, making this role essential.



Digital health specialist: Identified by 33% of respondents, the Digital Health Specialist will play a key role in integrating and managing digital health technologies within healthcare systems. This position will be crucial as the sector continues to adopt more sophisticated digital tools.



Telemedicine coordinator: With 33% of respondents foreseeing the growth of telehealth services, the role of telemedicine coordinator is expected to become increasingly important for managing remote patient care.



AI healthcare ethicist: As AI integration in healthcare expands, 33% of respondents believed there will be a need for professionals who can navigate the ethical challenges associated with AI applications, ensuring that these technologies are used responsibly.

Impact of Gen AI, AI, sector-specific technology, international mobility, DPI, climate, and green jobs on the job and skilling ecosystem



Artificial Intelligence (AI) and Gen AI: AI is expected to have a profound impact on the healthcare sector, with 66% of respondents rating its future impact as high as 9 or 10 on a scale of 10. However, respondents anticipate significant challenges in integration, including technical complexity, high costs of implementation and workforce resistance. 67% of the respondents expect AI to reshape job roles through reskilling and upskilling initiatives.



Sector-specific technology: Adoption of Industry 4.0 technologies is widespread, with 86% of respondents using IoT and 57% utilizing AI and Machine Learning. Despite this, the organizations feel there is lot of scope to be able to integrate AI and Gen AI, completely.

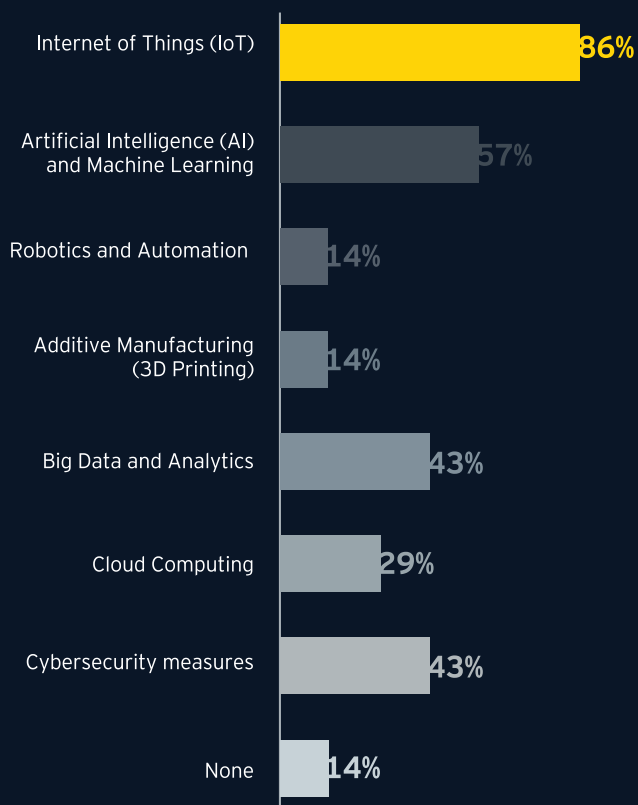


International mobility: Given that 71% of the organizations operate on an international scale, there is an increasing need for skills related to global operations. Cross-cultural competency and adaptability (50%) and global regulatory awareness (67%) are identified as crucial skills for the future healthcare workforce. Multilingual communication abilities are also highlighted by 33% of respondents as important for facilitating international mobility.

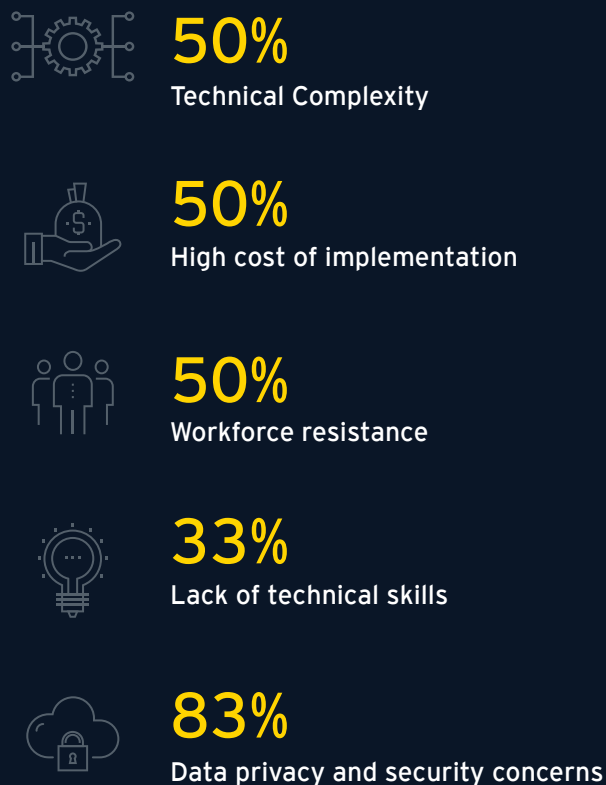


Climate and green jobs: Sustainability is a growing priority, with 43% of respondents identifying it as a top priority and another 43% considering it somewhat of a priority. However, 43% of organizations have not yet adopted any sustainable practices. Nevertheless, initiatives like waste reduction and recycling (67%) and increased energy efficiency (50%) are already underway in many organizations, highlighting a shift towards greener operations. On the jobs front, 67% of respondents affirm that economic feasibility and resistance to change are the primary challenges in creating and filling green jobs.

Utilization of any of the Industry 4.0 technologies



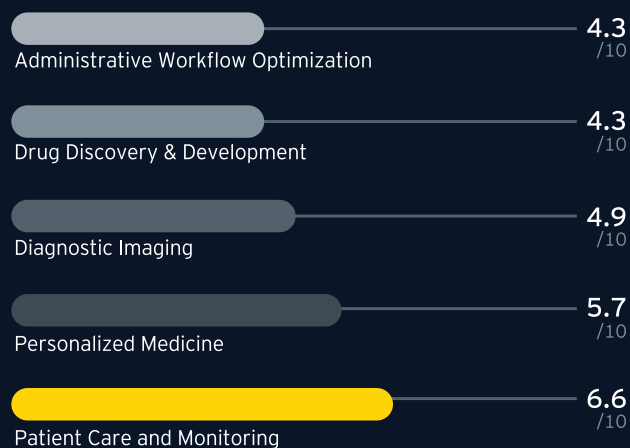
Challenges anticipated in integrating AI into existing Healthcare systems



How prepared are organizations to adapt AI and Gen AI



Expected Healthcare sectors to have greatest impact of AI



Suggested programmatic and policy interventions



For industry leaders

- ▶ **Reskilling and upskilling initiatives:** Industry leaders should prioritize regular training programs, as 67% of respondents emphasize the need for reskilling and upskilling to adapt to AI and other emerging technologies. These initiatives will be essential to address the gaps in technical skills and prepare the workforce for future challenges as industry 4.0 technologies take precedence.
- ▶ **Adopting sustainable practices:** With 43% of organizations already recognizing sustainability as a priority, it is critical to continue integrating sustainable practices into core operations. Green jobs, supported by targeted training programs, will be vital for this transition.
- ▶ **Enhanced collaboration with government:** To leverage government incentives effectively, industry leaders should engage in partnerships that align with policy objectives. These collaborations will help maximize the benefits of subsidies and incentives for technology adoption and sustainability.

How effective are current government policies in promoting technology adoption in the Healthcare sector

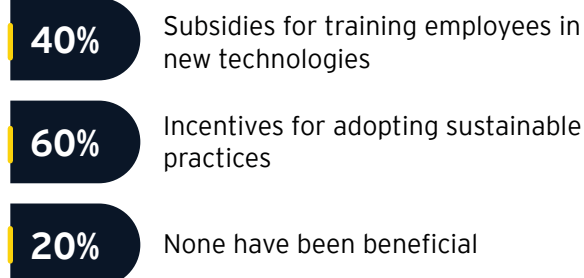


For government

- ▶ **Financial incentives for green investments:** With 60% of respondents indicating the need for financial incentives, the government should consider offering robust support to encourage the adoption of sustainable practices in healthcare.
- ▶ **Support for AI and technology integration:** The government can provide subsidies for employee training in new technologies, as 40% of respondents have identified this as beneficial. Stricter environmental regulations (40%) and incentives for adopting sustainable practices (60%) are also necessary to drive progress.
- ▶ **Policy support for workforce mobility and gender diversity:** To ensure an inclusive and diverse workforce, the government should consider implementing gender quotas (40%), offering grants for diversity initiatives (20%), and providing childcare support services (20%). Additionally, the government should play a supportive role (20%) in addressing skill gaps through public-private partnerships, ensuring that the healthcare sector is equipped to meet future demands.

These interventions will be critical in shaping the future of jobs in the healthcare sector, ensuring that the workforce is equipped with the necessary skills to adapt to technological advancements, sustainability challenges, and global operations

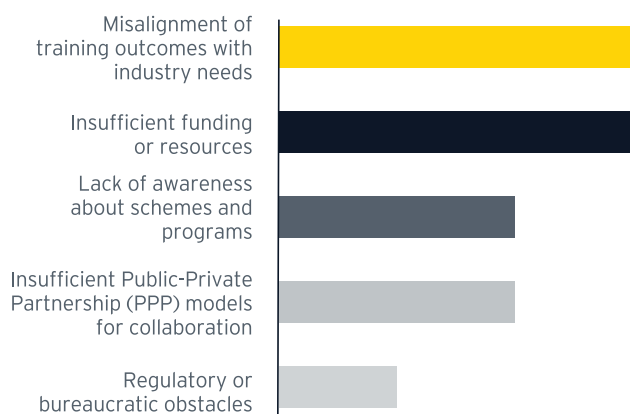
Government incentives/subsidies can be/ have been most beneficial for technological advancements



Areas skilling institutions should focus on



Challenges healthcare companies face in collaborating with educational and skilling institutions



3.2.5

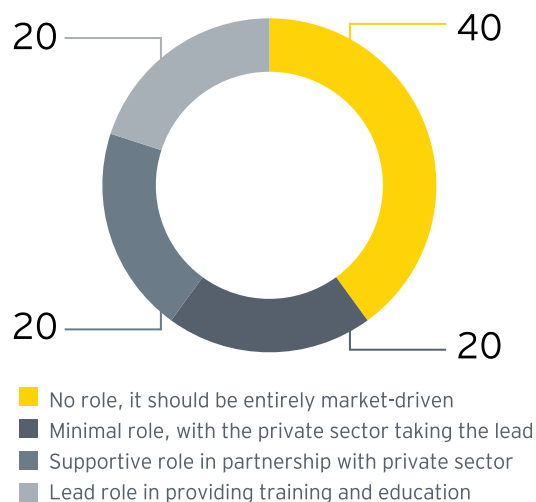
Overall impact on job roles and skills in the sector

Our primary and secondary research in the healthcare field shows encouraging trends. Both types of research emphasize the importance of healthcare to India's economy. They consistently predict solid growth and the creation of new jobs due to technological advancements. Primary research highlights the emergence of new roles like Health Data Analysts, Genomic Counsellors, and Digital Health Specialists. According to 50% of respondents, these roles will be crucial in the future.

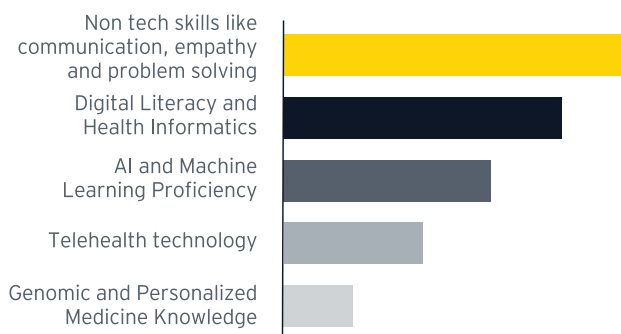
Emerging job roles in healthcare sector



Role should the government play in addressing skill gaps in the Healthcare sector



Skill sets that will be most in demand for future healthcare job roles



Full form of above mentioned variable "Non tech skills like communication , empathy and problem solving"

Collectively the research highlights the significant role of AI and emerging technologies in improving diagnosis and tailoring treatments. Research shows that 66% of respondents expect AI to significantly impact healthcare, although challenges such as technical complexity and resistance from the workforce remain. In the next few years, the healthcare sector will merge advanced technologies with traditional practices, creating exciting opportunities for those who can adapt and acquire the skills needed for the digital health era.



3.3 Infrastructure

The infrastructure sector in India plays a pivotal role in the economic progress of the country, propelling its overall development. This sector encompasses a wide array of services including transportation (roads, railways, airports, and ports), energy and electricity, real estate and construction, urban infrastructure (water supply, sanitation, and waste management), and telecommunications. The Indian government has been actively promoting infrastructure growth through initiatives like the National Infrastructure Pipeline (NIP) and the Smart Cities Mission, aiming to enhance the quality of life and attract foreign investment. The sector has seen substantial investments, both from public and private sources, leading to significant improvements and modernization. However, it faces certain challenges such as regulatory hurdles, financing gaps, and the need for technological upgrades. Despite these challenges, the infrastructure sector in India is on an upward trajectory, with ongoing projects set to bolster connectivity, improve efficiency, and key enabler in helping India become a US\$26 trillion economy.⁴⁷

3.3.1 Macroeconomic trends

Key statistics

The Indian infrastructure sector is on a promising trajectory



Forecasts indicate substantial growth from approximately US\$204.06 billion in 2024 to US\$322.27 billion by 2029



This growth reflects a robust Compound Annual Growth Rate (CAGR) of 9.57% over this period⁴⁸



The Budget for 2024-25 has further reinforced this momentum with an 11.1% increase in capital investment outlay for infrastructure, totaling INR11.11 lakh crore (approximately US\$133.86 billion), which is 3.4% of India's GDP



Additionally, the budget allocation for the Railways has seen a notable 5.8% increase, reaching INR2.55 lakh crore (US\$30.72 billion) compared to the previous year⁴⁹



The infrastructure sector's contribution to the GDP is anticipated to remain significant, accounting for approximately 3.4% of India's GDP in the fiscal year 2024-25⁵⁰. This underscores the sector's pivotal role in driving the country's economic progress.

Foreign Direct Investment (FDI) continues to be a strong pillar for the sector, with total investments in construction development and infrastructure reaching approximately US\$ 60.06 billion from April 2000 to December 2023. Of this, US\$26.54 billion was directed towards construction development, while US\$33.52 billion was invested in infrastructure.

Looking ahead, India is committed to improving its Logistics Performance Index ranking to 25 and reducing logistics costs from 14% to 8% of GDP over the next five years. These ambitious goals highlight the government's strategic focus on enhancing efficiency and competitiveness in the sector.



B Employment scenario

The infrastructure sector is a vital contributor to India's employment landscape with an estimation of 9.8 million jobs to be created in FY25⁵¹. This underscores the sector's critical role in providing job opportunities across the country.

In the construction sector, the percentage of workers has increased to 13.0% in 2022-23 as compared to 12.1% in 2020-21⁵². This scenario opens significant potential for new entrants into the workforce, particularly as the sector continues to expand.

The Union Budget for 2024-25 has reinforced this commitment by announcing plans to upgrade 1,000 ITIs using a hub-and-spoke model⁵³. This initiative is designed to align course content with current industry skill requirements and introduce new, relevant courses, thereby enhancing the quality of technical education and ensuring that graduates are well-prepared for careers in the infrastructure sector.

A new centrally sponsored scheme under the Prime Minister's package, set to be introduced in collaboration with state governments and industry partners, aims to skill 20 lakh first timers over the next five years. This ambitious initiative underscores the government's strong commitment to bridging the skill gap and ensuring a steady supply of skilled labor for the infrastructure sector.

While there is a recognized need for skilled labor, particularly certified professionals and those with specialized skills in operating advanced construction equipment and technologies, this also creates a promising landscape for upskilling and training initiatives. Addressing this skilled labor requirement is poised to unlock even greater potential in the sector, contributing to its sustained growth and the overall economic development of the country.

C Key drivers of growth and government schemes

Government of India has made a substantial commitment to infrastructure development, significantly increasing the budget allocation to INR5 lakh crore (approximately US\$60 billion) for FY2024, up from INR3.7 lakh crore in FY2023. This investment aligns with the objectives of the National Infrastructure Pipeline (NIP), which oversees projects worth INR108 trillion (approximately US\$1.3 trillion) at various stages of implementation, all aimed at enhancing the country's infrastructure.

The sector is also embracing emerging technologies such as smart construction, digital infrastructure, and renewable energy solutions, which are set to drive both efficiency and sustainability in infrastructure development.

Several major government initiatives are in place to support sustainable infrastructure development:

- ▶ **National Infrastructure Pipeline (NIP):** Focused on improving infrastructure quality while promoting green practices, the NIP plays a crucial role in the nation's long-term growth strategy.
- ▶ **PM Gati Shakti:** This initiative is accelerating the construction of national highways, with a target to complete 200,000 km by 2025, significantly enhancing connectivity.
- ▶ **UDAN Scheme:** Designed to improve air connectivity to underserved regions, the UDAN Scheme has operationalized 415 routes and 66 airports, fostering regional development.
- ▶ **Sagarmala Project:** Aimed at enhancing port connectivity and logistics, the Sagarmala Project has attracted significant investments, boosting cargo handling capacity and overall maritime infrastructure.

- **India Infrastructure Project Development Fund (IIPDF):** Supporting Public-Private Partnership (PPP) projects, the IIPDF provides financial assistance for project development expenses, encouraging private sector participation in infrastructure projects.
- The government is proactively addressing these areas to ensure sustained progress. The establishment of the Urban Infrastructure Development Fund (UIDF) with an annual outlay of INR10,000 crores is a testament to these efforts, targeting the improvement of urban infrastructure and quality of life in cities.
- To further bolster infrastructure development, the government has introduced diverse financial instruments aimed at mobilizing substantial resources for various projects. These include Infrastructure Debt Funds (IDFs), Real Estate Investment Trusts (REITs), and tax-free bonds, all contributing to a more dynamic and resilient financing ecosystem.

Collectively, these comprehensive policies and initiatives underscore the government's strong commitment to advancing India's infrastructure landscape, overcoming existing challenges, and paving the way for sustained economic prosperity.

3.3.2 International benchmark

India has achieved remarkable progress in its infrastructure sector, as evidenced by its improved ranking in the World Bank's Logistics Performance Index (LPI). In 2023, India advanced to 38th place out of 139 countries, up from 44th in 2018. This significant improvement reflects the effectiveness of key government initiatives such as PM Gati Shakti and the National Logistics Policy, both of which are focused on enhancing the nation's soft and hard infrastructure. Furthermore, India has made substantial progress in the International Shipments category, climbing from 44th in 2014 to 22nd in 2023, driven by successful modernization and digitalization efforts within the logistics sector⁵⁴.

In addressing job roles and skills within the infrastructure sector, international best practices offer valuable insights:



Long-term strategic planning: Countries like the UK and Australia emphasize the importance of long-term strategic infrastructure planning, which aligns workforce skills with future needs, ensuring the sector remains adaptable and prepared for upcoming challenges⁵⁵.



Skill development roadmaps: The UK's National Infrastructure Commission provides a comprehensive roadmap for future skill requirements, allowing educational institutions to tailor their curricula to meet these evolving demands.



AI and efficiency: Germany and the USA are at the forefront of leveraging artificial intelligence (AI) to optimize construction processes and project management, enhancing efficiency and reducing costs.



Sustainable practices: The Netherlands and Scandinavian countries are leaders in integrating green building technologies and renewable energy into infrastructure projects. This focus on sustainability is driving the development of green skills through specialized training programs, creating new job roles such as sustainability consultants and green building specialists.

India's continued emphasis on adopting such best practices, coupled with its commitment to modernization and sustainability, positions the nation to further enhance its infrastructure sector while simultaneously fostering the development of a skilled, future-ready workforce.

3.3.3 Sectoral trends

Emerging technologies in the sector

The infrastructure sector is undergoing a dynamic transformation, driven by the adoption of Industry 4.0 technologies. These key emerging technologies are setting the stage for enhanced efficiency, safety, and sustainability across various infrastructure projects:

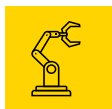


Internet of Things (IoT): The IoT facilitates a network of interconnected devices and sensors that collect and exchange data, enabling real-time monitoring and decision-making in infrastructure projects. For example, real-time monitoring allows for the early identification of potential issues, enabling proactive maintenance and minimizing downtime.



Big data analytics: By analyzing large sets of data, big data analytics enhances forecasting, risk management, and operational efficiency, leading to more

informed decisions in infrastructure development.



Autonomous robots and drones: These technologies are increasingly being used for surveying, inspection, and maintenance tasks, improving safety and efficiency on construction sites.

In ports, digitalization is revolutionizing operations and boosting efficiency, while in the power sector, AI and machine learning are optimizing energy generation and grid security. The transportation sector is benefiting from AI systems that enhance road safety and monitor infrastructure conditions, and water management is seeing improvements through real-time monitoring and predictive analytics. In the oil and gas industry, digital communication technologies are enhancing asset management and predictive maintenance.

While the integration of these advanced technologies presents challenges, such as the need for a skilled workforce and the implementation of robust cybersecurity measures, these are also opportunities for growth and innovation. As India continues to invest in its infrastructure, fostering public-private partnerships and enhancing digital connectivity will be pivotal in driving these advancements forward, ultimately improving the quality of life for its citizens.

B Workforce trends

According to the National Skill Development Council (NSDC), the real estate sector is a major employer, which includes both skilled and unskilled labor. The

infrastructure sector complements this by employing a large section of the workforce. Among the 71 million individuals working in the construction industry, there are approximately 4.4 million core skilled workers, including engineers, technicians, and clerical staff, alongside 6.9 million employees who have received vocational training.

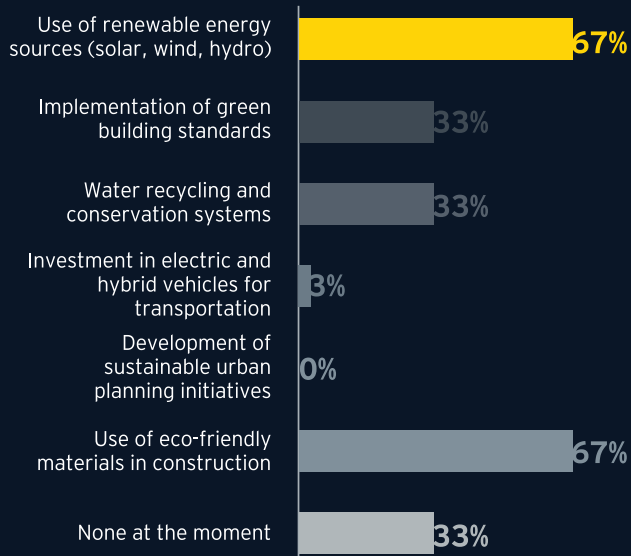
The demand for freshers across various sectors has increased by 5% over the past six months, signaling a positive recovery in hiring trends. This uptrend is bolstered by improvements in the IT sector, which also has a beneficial impact on infrastructure-related roles. As the industry evolves, specialized certifications such as Project Management Professional (PMP) and LEED accreditation are becoming increasingly valuable for roles focused on sustainability and project delivery.

While there are challenges in workforce development, particularly in bridging the skill gap between current capabilities and the needs of modern infrastructure projects, these challenges also present opportunities for growth. The gap is especially pronounced in areas related to sustainable practices and advanced technologies, where there is significant potential for upskilling. The National Federation of Independent Business (NFIB) has noted that 43% of business owners report job openings that are difficult to fill, underscoring the need for targeted training and development programs.

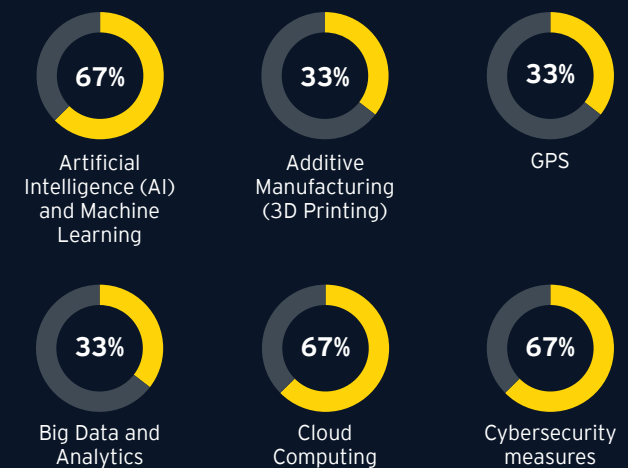
Retention remains an area for improvement, as high turnover rates can disrupt project continuity and increase training costs. However, addressing these challenges with strategic workforce development initiatives will enhance the sector's resilience and contribute to its ongoing success.



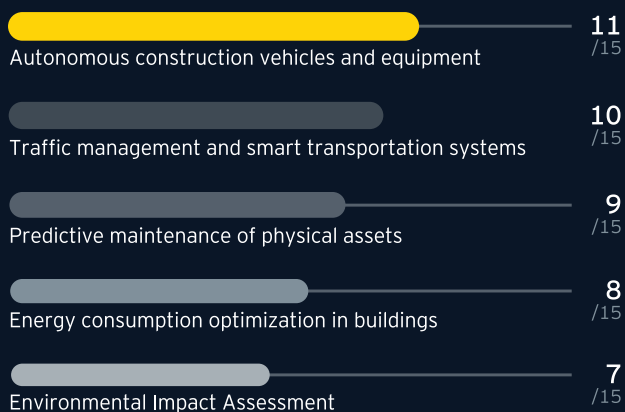
Sustainable practices the organizations in the sector have adopted



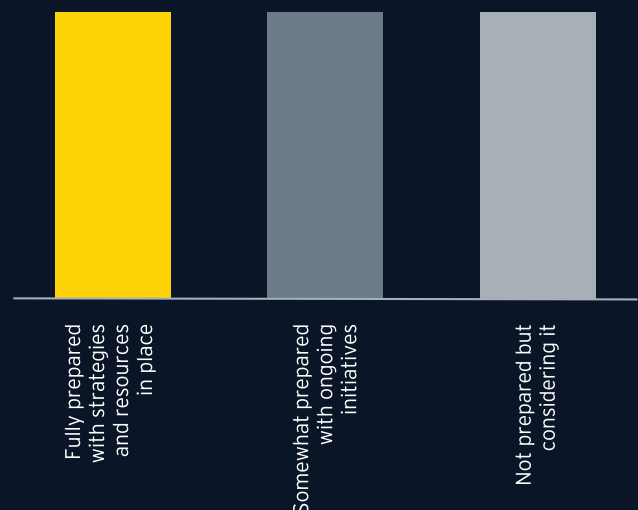
Utilization of any of the Industry 4.0 technologies



Areas of infrastructure sector where AI is expected to have the greatest impact



Preparedness level of organizations to adopt AI and Gen AI technologies



Biggest challenge in workforce development in infrastructure operations

